

Zara- E-commerce management system

Seller and Inventory Management

1.Create seller and inventory table

1.1 Seller table:

Attribute	Description	Key
Seller_ID	Unique identification number of the seller	Primary Key (PK)
Seller_Name	Name of the seller	—
Contact_No	Contact number of the seller	—
Email	Email address of the seller	—

1.2 Inventory Table:

Attribute	Description	Key
Inventory_ID	Unique identification number of the inventory record	Primary Key (PK)
Seller_ID	Identifies the seller associated with the inventory	Foreign Key (FK)
Product_ID	Identifies the product in the inventory	Foreign Key (FK)
Stock_Quantity	Number of units currently available in stock	—
Stock_Status	Indicates whether the product is available or unavailable	—

1.3 SQL code for Seller table

```
SQL> CREATE TABLE Seller (  
2     Seller_ID INT PRIMARY KEY,  
3     Seller_Name VARCHAR(50),  
4     Contact_No VARCHAR(15),  
5     Email VARCHAR(50)  
6 );  
  
Table created.
```

1.4 SQL code for Inventory table

```
SQL> CREATE TABLE Inventory (  
2     Inventory_ID INT PRIMARY KEY,  
3     Seller_ID INT,  
4     Product_ID INT,  
5     Stock_Quantity INT,  
6     Stock_Status VARCHAR(20),  
7     FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID),  
8     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)  
9 );  
  
Table created.
```

2.Inserting values

2.1 Insert 5 seller records

```
SQL> INSERT INTO Seller VALUES  
2 (1, 'Zara India', '9876543210', 'zaraindia@gmail.com');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller VALUES  
2 (2, 'Fashion Hub', '9876543211', 'fashionhub@gmail.com');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller VALUES  
2 (3, 'Style Store', '9876543212', 'stylestore@gmail.com');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller VALUES  
2 (4, 'Trend Wear', '9876543213', 'trendwear@gmail.com');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller VALUES  
2 (5, 'Urban Fashion', '9876543214', 'urbanfashion@gmail.com');  
  
1 row created.
```

2.2 Insert 5 Inventory records

```

SQL> INSERT INTO Inventory VALUES
  2 (1001, 1, 101, 25, 'Available');

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES
  2 (1002, 2, 102, 0, 'Unavailable');

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES
  2 (1003, 3, 103, 15, 'Available');

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES
  2 (1004, 4, 104, 0, 'Unavailable');

1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES
  2 (1005, 5, 105, 30, 'Available');

1 row created.

```

3. Establish Relationships Between Sellers, Products and Stock

The database establishes relationships between SELLER, PRODUCT, and INVENTORY using primary keys and foreign keys.

- SELLER → PRODUCT: One seller can provide multiple products.
- PRODUCT → INVENTORY: A product is connected to its inventory record.
- SELLER_ID in PRODUCT is a foreign key referencing SELLER.
- PRODUCT_ID in INVENTORY is a foreign key referencing PRODUCT.

Relationship Structure:

SELLER

|

| Supplies

↓

PRODUCT

|

| Has Stock

↓

INVENTORY

4. Maintain Seller product information

```
SQL> ALTER TABLE PRODUCT
  2  ADD SELLER_ID NUMBER;

Table altered.

SQL> ALTER TABLE PRODUCT
  2  ADD CONSTRAINT FK_PRODUCT_SELLER
  3  FOREIGN KEY (SELLER_ID)
  4  REFERENCES SELLER(SELLER_ID);

Table altered.
```

5. Assign sellers to products

```
SQL> UPDATE PRODUCT
  2  SET SELLER_ID = 1
  3  WHERE PRODUCT_ID = 101;

1 row updated.

SQL>
SQL> UPDATE PRODUCT
  2  SET SELLER_ID = 2
  3  WHERE PRODUCT_ID = 102;

1 row updated.

SQL>
SQL> UPDATE PRODUCT
  2  SET SELLER_ID = 3
  3  WHERE PRODUCT_ID = 103;

1 row updated.

SQL>
SQL> UPDATE PRODUCT
  2  SET SELLER_ID = 4
  3  WHERE PRODUCT_ID = 104;

1 row updated.

SQL>
SQL> UPDATE PRODUCT
  2  SET SELLER_ID = 5
  3  WHERE PRODUCT_ID = 105;

1 row updated.

SQL>
SQL> COMMIT;
```

6. Display Seller Product Information

```
SQL> SELECT
  2     S.SELLER_ID,
  3     S.SELLER_NAME,
  4     P.PRODUCT_ID,
  5     P.PRODUCT_NAME
  6 FROM SELLER S
  7 JOIN PRODUCT P
  8 ON S.SELLER_ID = P.SELLER_ID;
```

SELLER_ID	SELLER_NAME	PRODUCT_ID	PRODUCT_NAME
1	Zara India	101	Floral Dress
2	Fashion Hub	102	Casual T-Shirt
3	Style Store	103	Slim Fit Jeans
4	Trend Wear	104	Running Shoes
5	Urban Fashion	105	Leather Belt

7. Track Available and Unavailable Products

```
SQL> SELECT
  2     I.INVENTORY_ID,
  3     I.PRODUCT_ID,
  4     P.PRODUCT_NAME,
  5     I.STOCK_QUANTITY,
  6     CASE
  7         WHEN I.STOCK_QUANTITY > 0 THEN 'Available'
  8         ELSE 'Unavailable'
  9     END AS PRODUCT_STATUS
 10 FROM INVENTORY I
 11 JOIN PRODUCT P
 12 ON I.PRODUCT_ID = P.PRODUCT_ID;
```

INVENTORY_ID	PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY	PRODUCT_STATUS
1001	101	Floral Dress	25	Available
1002	102	Casual T-Shirt	0	Unavailable
1003	103	Slim Fit Jeans	15	Available
1004	104	Running Shoes		

INVENTORY_ID	PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY	PRODUCT_STATUS
1001	101	Floral Dress	25	Available
1002	102	Casual T-Shirt	0	Unavailable
1003	103	Slim Fit Jeans	15	Available
1004	104	Running Shoes	0	Unavailable
1005	105	Leather Belt	30	Available

8.Generate Inventory Status Report

The inventory status report displays the seller name, product details, available stock, and product availability. It helps identify whether each product is Available or Unavailable based on its stock quantity.

```
SQL> SELECT
  2     S.SELLER_NAME,
  3     P.PRODUCT_ID,
  4     P.PRODUCT_NAME,
  5     I.STOCK_QUANTITY,
  6     CASE
  7         WHEN I.STOCK_QUANTITY > 0 THEN 'Available'
  8         ELSE 'Unavailable'
  9     END AS PRODUCT_STATUS
 10 FROM SELLER S
 11 JOIN PRODUCT P
 12 ON S.SELLER_ID = P.SELLER_ID
 13 JOIN INVENTORY I
 14 ON P.PRODUCT_ID = I.PRODUCT_ID;
```

```
SELLER_NAME                                PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STA
-----
Zara India                                101
Floral Dress
      25 Available

Fashion Hub                                102
Casual T-Shirt
      0 Unavailable
```

```

SELLER_NAME                                     PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY  PRODUCT_STA
-----

```

```

Style Store                                     103
Slim Fit Jeans
      15 Available

```

```

Trend Wear                                     104
Running Shoes

```

```

SELLER_NAME                                     PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY  PRODUCT_STA
-----

```

```

      0 Unavailable

```

```

Urban Fashion                                     105
Leather Belt
      30 Available

```

9. PL/SQL Inventory Status Report

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
2
3     CURSOR inventory_cursor IS
4     SELECT
5         S.SELLER_NAME,
6         P.PRODUCT_ID,
7         P.PRODUCT_NAME,
8         I.STOCK_QUANTITY
9     FROM SELLER S
10    JOIN PRODUCT P
11    ON S.SELLER_ID = P.SELLER_ID
12    JOIN INVENTORY I
13    ON P.PRODUCT_ID = I.PRODUCT_ID;
14
15     v_seller_name SELLER.SELLER_NAME%TYPE;
16     v_product_id PRODUCT.PRODUCT_ID%TYPE;
17     v_product_name PRODUCT.PRODUCT_NAME%TYPE;
18     v_stock INVENTORY.STOCK_QUANTITY%TYPE;
19
20 BEGIN
21
22     OPEN inventory_cursor;
23
24     LOOP
25
26         FETCH inventory_cursor
27         INTO v_seller_name,
28             v_product_id,
29             v_product_name,
30             v_stock;
31
32         EXIT WHEN inventory_cursor%NOTFOUND;

```

```

33
34     DBMS_OUTPUT.PUT_LINE('Seller Name: ' || v_seller_name);
35     DBMS_OUTPUT.PUT_LINE('Product ID: ' || v_product_id);
36     DBMS_OUTPUT.PUT_LINE('Product Name: ' || v_product_name);
37     DBMS_OUTPUT.PUT_LINE('Stock Quantity: ' || v_stock);
38
39     IF v_stock > 0 THEN
40         DBMS_OUTPUT.PUT_LINE('Product Status: Available');
41     ELSE
42         DBMS_OUTPUT.PUT_LINE('Product Status: Unavailable');
43     END IF;
44
45     DBMS_OUTPUT.PUT_LINE('-----');
46
47 END LOOP;
48
49 CLOSE inventory_cursor;
50
51 END;
52 /

```

```

Seller Name: Zara India
Product ID: 101
Product Name: Floral Dress
Stock Quantity: 25
Product Status: Available
-----

```

```

Seller Name: Fashion Hub
Product ID: 102
Product Name: Casual T-Shirt
Stock Quantity: 0
Product Status: Unavailable
-----

```

```

Seller Name: Style Store
Product ID: 103
Product Name: Slim Fit Jeans
Stock Quantity: 15
Product Status: Available
-----

```

```

Seller Name: Trend Wear
Product ID: 104
Product Name: Running Shoes
Stock Quantity: 0
Product Status: Unavailable
-----

```

```

Seller Name: Urban Fashion
Product ID: 105
Product Name: Leather Belt
Stock Quantity: 30
Product Status: Available
-----

```

PL/SQL procedure successfully completed.